



Making Maternal Child Health Data Visuals in R

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What We'll Cover

- Why we visualize, and the **ggplot recipe**
- Building a simple fake **MCH dataset**
- Six charts: **bar, line, boxplot, scatter, lollipop, facets**
- Polishing with **labels & themes**
- Publication tables with the **gt** package
- A short **hands-on exercise** — your turn



PART ONE

The ggplot Recipe



Every Chart Is the Same Three Parts

- **1 · DATA** — the table you want to show
- **2 · aes()** — which column maps to x, y, color...
- **3 · geom_** — how to draw it: bars, lines, points...
- Read **+** as “add a layer”; swap the geom for a new chart

```
ggplot(data, aes(x = state, y = rate)) +  
  geom_col()      # swap this line to change the chart type
```



Our Fake MCH Dataset

- One row per **county** × **year** (2015–2023)
- Idaho, Kansas & Montana counties
- **pct_lbw** — % low birth weight
- **pct_preterm** — % preterm birth
- **pct_prenatal** — % adequate prenatal care
- Real work: **births <- read_csv("file.csv")**

```
library(tidyverse)

# view what your data looks like
glimpse(births)

# state   county   year pct_lbw pct_preterm
# Idaho   Ada      2023  6.7     8.4
# Kansas  Sedgwick 2023  7.9     9.8
# Montana Flathead 2023  7.0     8.9

births_2023 <- filter(births,
                      year == 2023)
```



PART TWO

Six Charts for MCH Data

Bar · Compare Groups — `geom_col()`

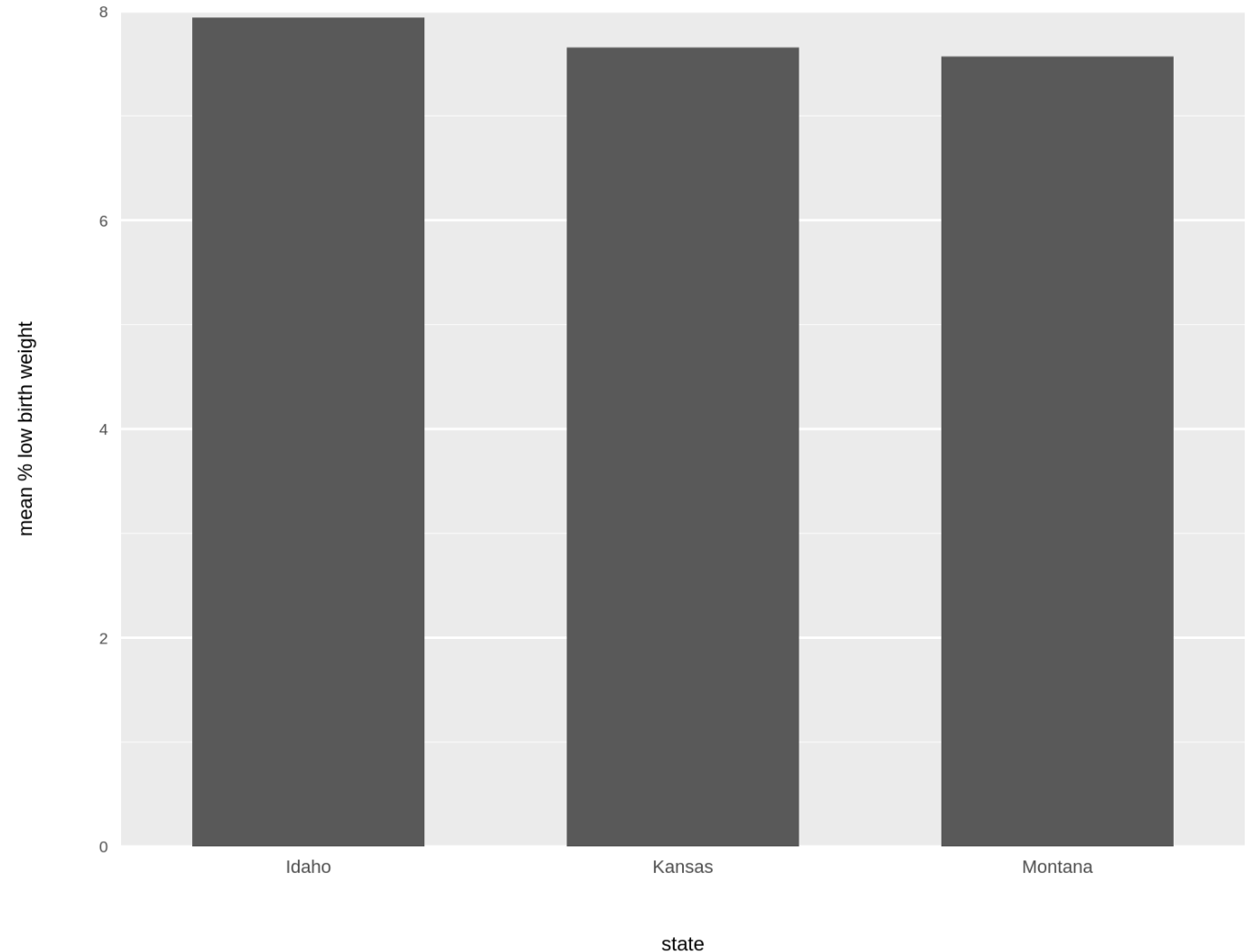
```
births_2023 |>
  group_by(state) |>
  summarise(
    lbw = mean(pct_lbw)) |>
  ggplot(aes(state, lbw)) +
  geom_col()

# |> is the base pipe

# %>% is the magrittr pipe

# ggplot uses "+" to step between each
ggplot layer
```

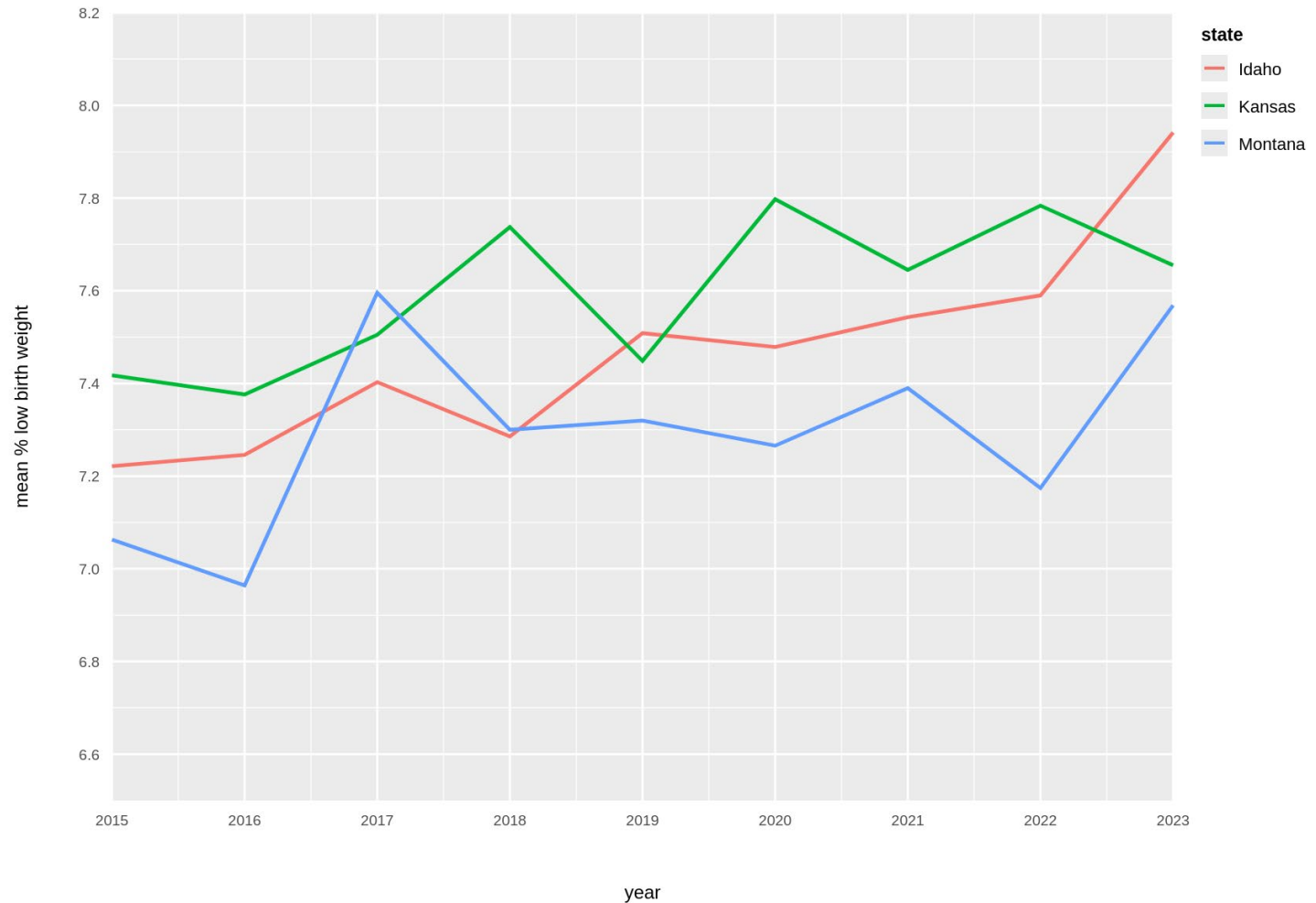
Heights come from a value you compute with **`summarise()`**. Additional information: [ggplot2 geom_bar\(\) vs geom_col\(\) in R: Bar Charts Made Easy](#)



Line · Trends Over Time — geom_line()

```
births |>
  group_by(state, year) |>
  summarise(
    lbw = mean(pct_lbw) ) |>
  ggplot(aes(year, lbw,
    color = state)) +
  geom_line(linewidth = 1)
```

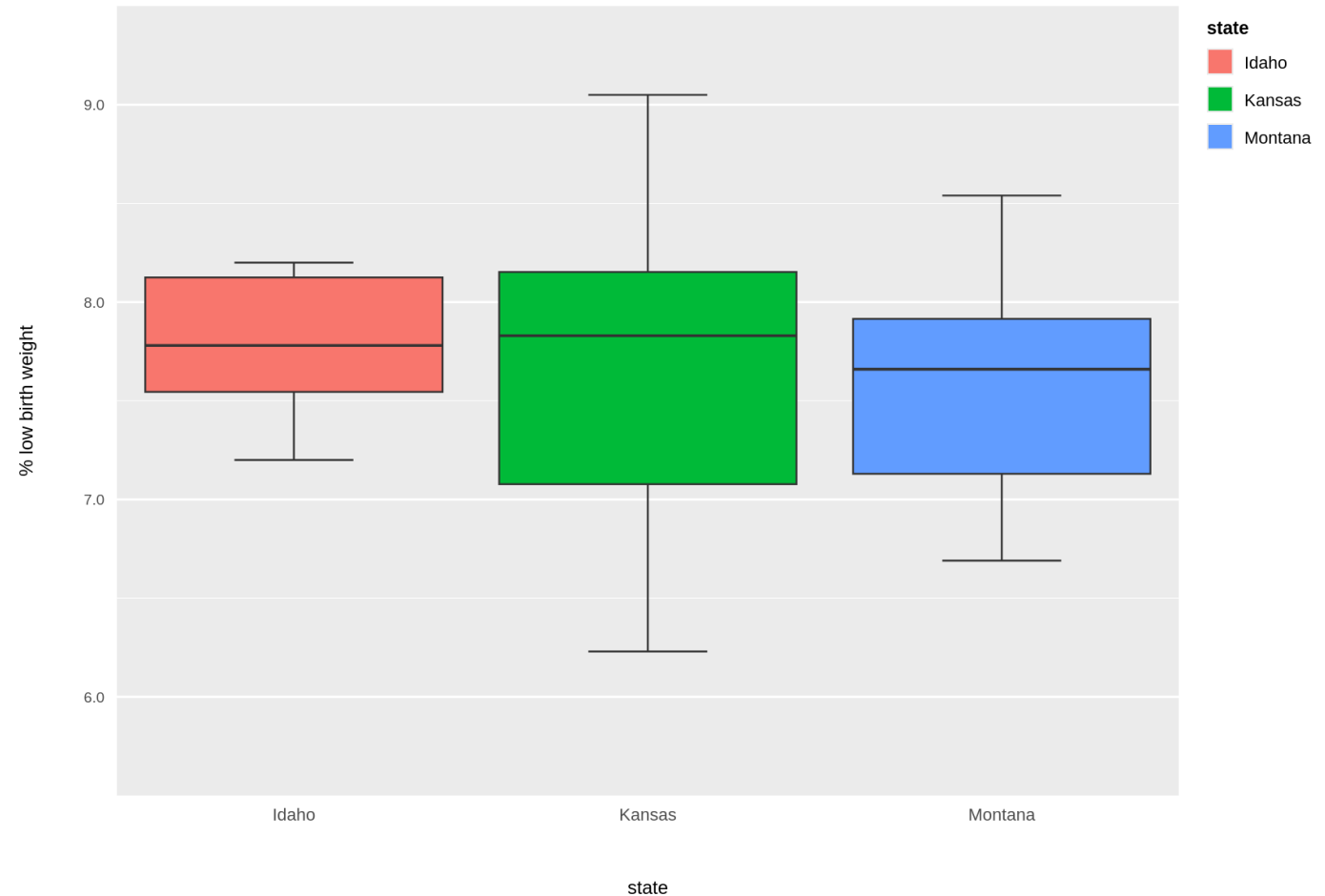
color = state makes one line per group — and a legend.



Distribution · Spread — geom_boxplot()

```
ggplot(births_2023,  
  aes(x = state,  
      y = pct_lbw,  
      fill = state)) +  
  geom_boxplot()
```

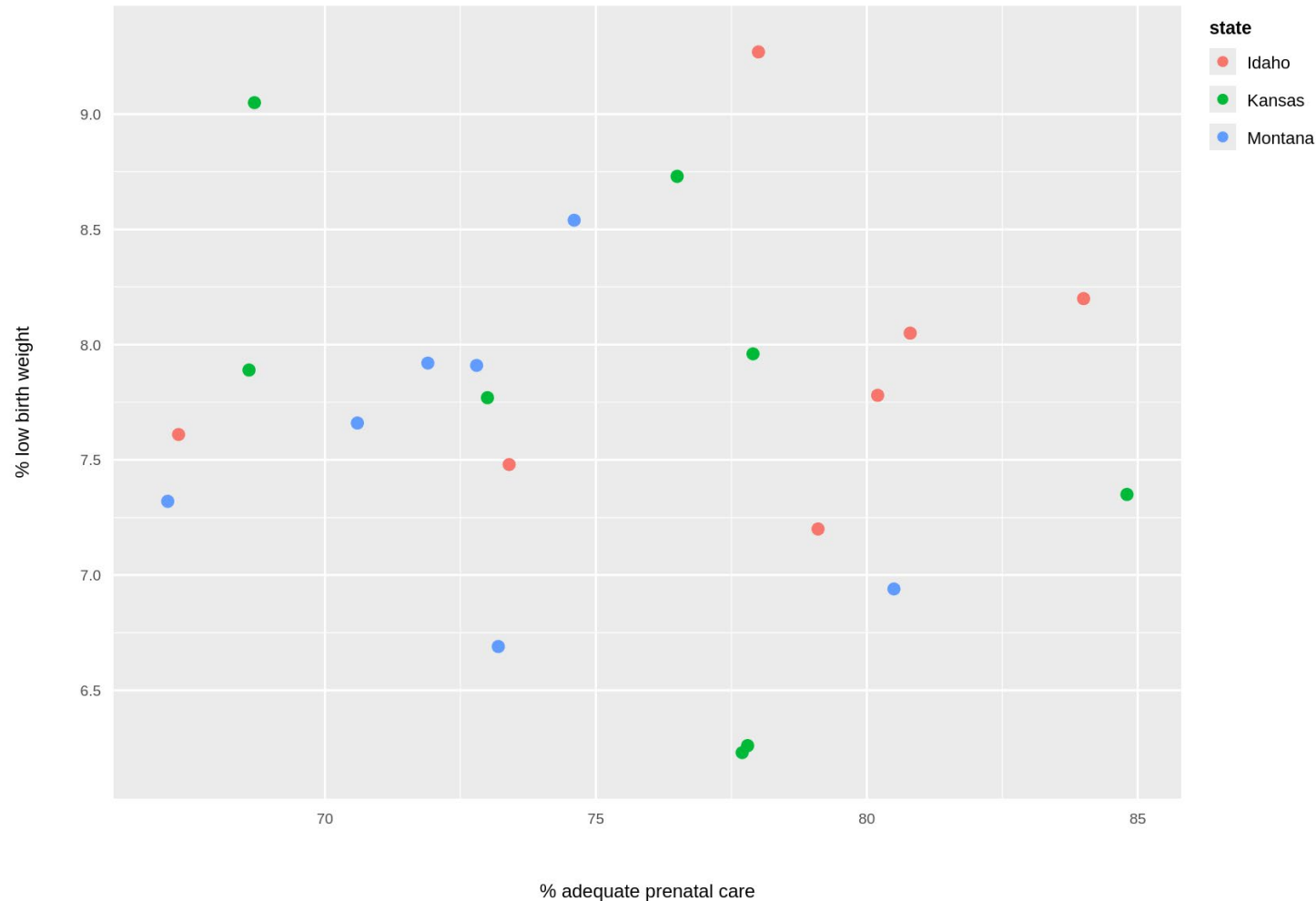
The box is the middle 50% of counties; line = median.



Scatter · Relationships — geom_point()

```
ggplot(births_2023,  
  aes(x = pct_prenatal,  
      y = pct_lbw,  
      color = state)) +  
  geom_point(size = 3)
```

Each point is one county. Do the two rates move together?

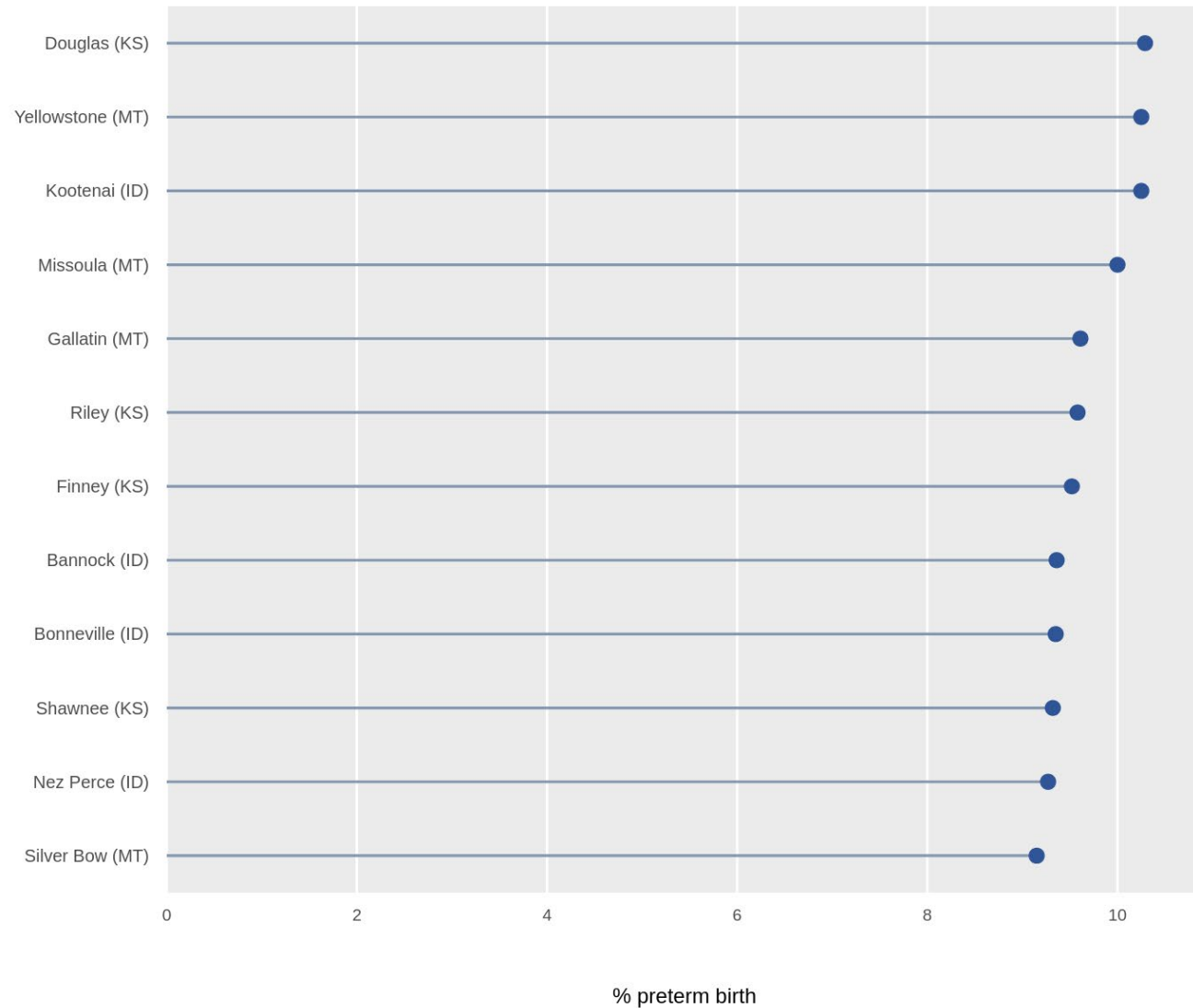


Lollipop · Ranking Counties — geom_segment()

```
births_2023 |>
  slice_max(pct_preterm, n = 12) |>
  ggplot(aes(pct_preterm,
    reorder(county,
      pct_preterm))) +
  geom_segment(aes(xend = 0,
    yend = county)) +
  geom_point(size = 3)
```

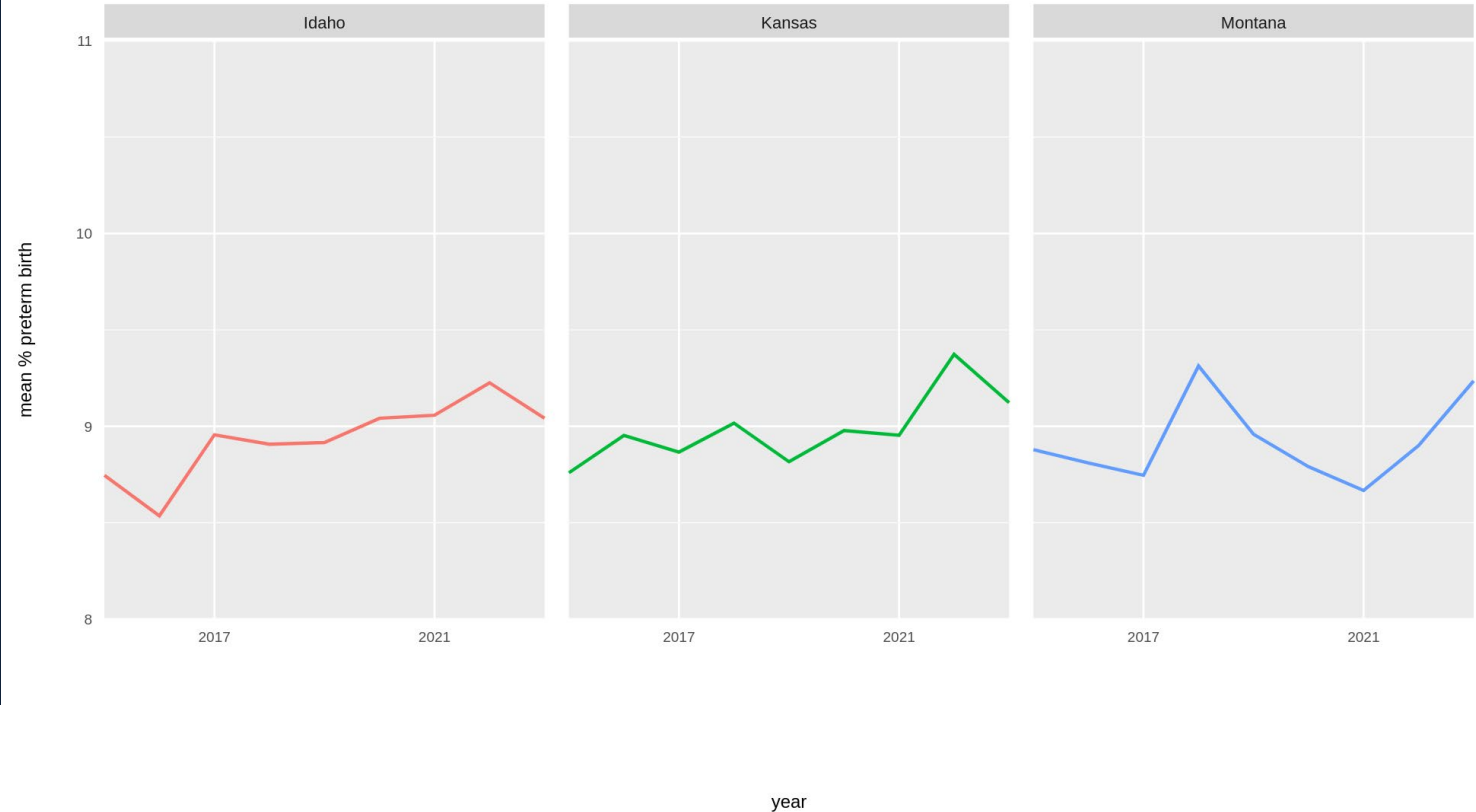
reorder() sorts counties by their rate.

county



Small Multiples — facet_wrap()

```
births |>
  group_by(state, year) |>
  summarise(preterm =
    mean(pct_preterm)) |>
  ggplot(aes(year, preterm,
    color = state)) +
  geom_line() +
  facet_wrap(~ state)
```



One panel per state, shared axes — easy to compare.



Make It Presentable

```
ggplot(births_2023,  
  aes(pct_prenatal, pct_lbw,  
      color = state)) +  
  geom_point(size = 3) +  
  labs(  
    title = "Prenatal care & LBW",  
    x = "% adequate prenatal care",  
    y = "% low birth weight",  
    color = "State") +  
  theme_minimal(base_size = 14)  
  
ggsave("plot.png", width = 8,  
  height = 5, dpi = 300)
```

- **labs()** — titles, axis names, legend, caption
- **theme_minimal()** — a clean look; also `theme_bw()`
- **ggsave()** — export to PNG for slides & reports



Nice Tables — The gt Package

```
library(gt)

births_2023 |>
  select(state, county, pct_lbw,
         pct_preterm, pct_prenatal) |>
  gt() |>
  tab_header(title =
    "County Birth Outcomes") |>
  fmt_number(starts_with("pct"),
    decimals = 1)
```

County Birth Outcomes, 2023

Selected counties across Idaho, Kansas, and Montana

State	County	% LBW	% Preterm	% Prenatal
Idaho	Ada	6.7	8.4	84.1
Idaho	Canyon	7.4	9.1	79.6
Kansas	Sedgwick	7.9	9.8	76.3
Kansas	Johnson	6.9	8.7	82.5
Montana	Yellowstone	7.3	9.2	74.8
Montana	Flathead	7.0	8.9	77.1

Source: Simulated MCH surveillance data (teaching example)

Uses the same tidyverse pipe style— exports to Word & PPT.



PART THREE

Your Turn



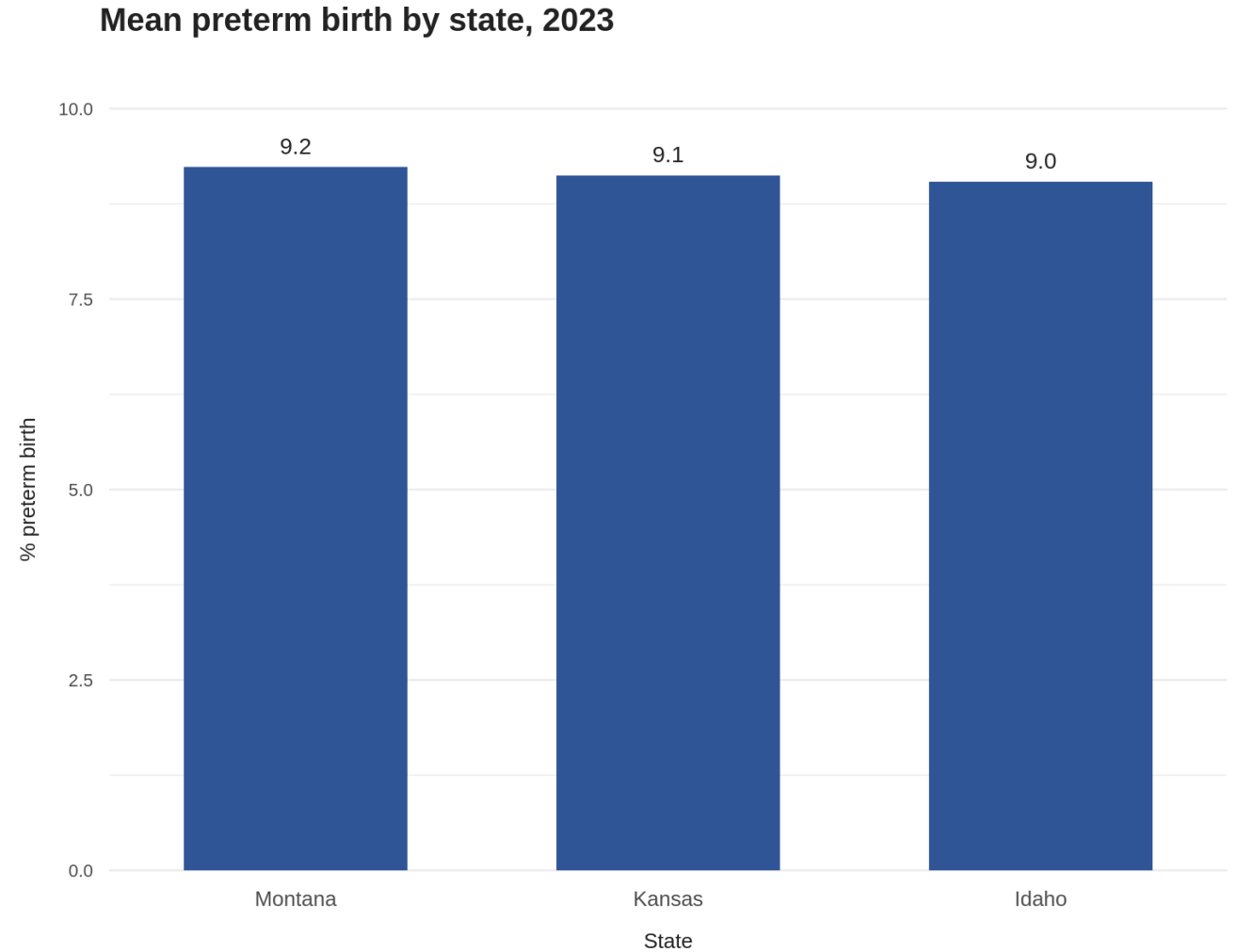
Exercise · Build Your Own Bar Chart

```
# Goal: mean % preterm by state,  
#       2023, with a real title.  
  
births_2023 |>  
  group_by(____) |>  
  summarise(preterm =  
    mean(____)) |>  
  ggplot(aes(state, ____)) +  
  geom_col(fill = "#2F5597") +  
  labs(title = "____")
```

- **Stretch goals — pick one:**
 - Sort bars high→low with `reorder(state, -preterm)`
 - Flip sideways: `+ coord_flip()`
 - Label bars: `+ geom_text(aes(label = round(preterm,1)))`

Exercise · One Solution

```
births_2023 |>
  group_by(state) |>
  summarise(preterm =
    mean(pct_preterm)) |>
  ggplot(aes(reorder(state, -preterm),
    preterm)) +
  geom_col(fill = "#2F5597") +
  geom_text(aes(label =
    round(preterm, 1)),
    vjust = -0.4) +
  labs(title = "Mean preterm birth by state",
    x = "State",
    y = "% preterm birth") +
  theme_minimal(base_size = 14)
```





Takeaways & Cheat Sheet

- One recipe: **data + aes() + geom_**
- Swap the **geom** to change the chart
- **color / fill = column** splits by group
- **facet_wrap()** for small multiples
- **labs() + theme_minimal()** to polish
- The companion **.R script** has it all included

```
geom_col()      bars
geom_line()     trends
geom_point()    scatter
geom_boxplot()  spread
geom_segment()  lollipop
facet_wrap(~x)  small multiples
labs()          theme_minimal()
ggsave()        export
```



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Thank You/Questions

